

Brunel Supply Chain Logistics Problem Dataset

Supply Chain Logistics Problem Dataset

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Original: brunel.figshare.com/articles/dataset/Supply_Chain_Logistics_Problem_Dataset/7558679

Mirror (Excel + CSVs): github.com/jaredbach/LogisticsDataset

Local Excel: [raw/sources/brunel-logistics-dataset/SupplyChainLogisticsProblems.xlsx](#)

Related paper: [doi:10.1016/j.cie.2020.106610](https://doi.org/10.1016/j.cie.2020.106610)

This document converts the Figshare dataset description and the Excel workbook schema into plain text (and PDF) for reading without opening Excel.

Purpose

Seven tables define a logistics **order-routing** / **plant-assignment** problem:

1. **OrderList** — historical orders that need (or historically had) a route.
2. Six constraint tables — couriers/rates, plant-port links, plant-product support, VMI customer locks, daily plant capacities, and per-unit storage costs.

OrderList is historical routing and demand satisfaction. The other tables are **current network constraints**. You can cost the historical network and re-optimize under the new constraints.

For a linear program, take from each order: **Product_ID**, **Dest_Port**, **Unit_Quant** (cost), and **Weight** (weight bands); enforce limits from the other tables.

Workbook inventory

Sheet	Rows (excl. header)	Columns	Role
OrderList	9,215	14	Historical orders / demand
FreightRates	1,540	11	Courier lanes, weight bands, rates
WhCosts	19	2	Storage cost per unit by plant
WhCapacities	19	2	Max orders/day by plant
ProductsPerPlant	2,036	2	Allowed plant-product pairs
VmiCustomers	14	2	VMI plant-customer exclusivity
PlantPorts	22	2	Allowed plant-port links

Official sheet descriptions (from Figshare)

OrderList

Customer orders that need a route assigned. Includes order ID, product ID, customer, unit quantity, item weight, and historical routing fields (origin port, carrier, plant, destination port, service level, ship-ahead/late

days, transport day count).

- **Order ID** — ID of the order made by the customer.
- **Product ID** — specific product the customer ordered.

Historical note: carrier **V44_3** appears in OrderList but not in FreightRates. It was used historically, then discontinued. All V44_3 rows are **CRF** (customer arranges own shipping), so freight cost is not calculated for them anyway.

FreightRates

Available couriers, weight gaps (bands) for each lane, and associated rates. **tpt_day_cnt** / **TPT_Day_Count** = transportation day count (estimated shipping time).

PlantPorts

Allowed links between warehouses (plants) and shipping ports in the real network.

ProductsPerPlant

Supported warehouse-product combinations.

VmiCustomers

Special cases: listed warehouses may only support the listed customers. Any other non-listed warehouse can supply any customer.

WhCapacities

Warehouse capacities as **number of orders per day** (not units, not product-specific).

Example: Customer 1 requests 10 units of X, Customer 2 requests 20 units of Y → **2 orders**, so capacity consumed is 2.

WhCosts

Cost of storing products in a given warehouse, in **dollars per unit**.

Column data dictionary

OrderList

Column	Type (observed)	Description
Order_ID	number (unique)	Order identifier
Order_Date	date	Order date (all rows 2013-05-26 in this extract)
Orig_Port	string	Origin shipping port (e.g. PORT04, PORT05, PORT09)
Carrier	string	Carrier code used historically (V444_0, V444_1, V44_3)
TPT_Day_Count	integer	Transport / transit day count
Service_Level	string	CRF (customer freight), DTD (door-to-door), DTP (door-to-port)
Ship_Ahead_Day_Count	integer	Days shipped ahead of plan
Ship_Late_Day_Count	integer	Days shipped late

Column	Type (observed)	Description
Customer	string	Customer ID
Product_ID	integer	Product ordered
Plant_Code	string	Fulfilling plant / warehouse
Dest_Port	string	Destination port (PORT09 only in this extract)
Unit_Quant	number	Units ordered (for storage/shipping cost)
Weight	number	Shipment weight (for freight weight bands)

Cardinality (full sheet): 9,215 orders; 46 customers; 772 products; 7 plants; 3 origin ports; 3 carriers; 3 service levels.

FreightRates

Column	Type (observed)	Description
Carrier	string	Courier / carrier code
Orig_Port	string	Lane origin port
Dest_Port	string	Lane destination port
Min_Weight_Quant	number	Inclusive lower bound of weight band
Max_Weight_Quant	number	Inclusive upper bound of weight band
Service_Level	string	DTD or DTP
Min_Cost	number	Minimum charge for the band
Rate	number	Per-weight rate for the band
Mode_DSC	string	Mode: AIR or GROUND
TPT_Day_Count	integer	Estimated shipping days for this lane/option
Carrier_Type	string	Carrier type category

Cardinality: 1,540 rate rows; 9 carriers (V444_0–2, V444_4–9); Dest_Port = PORT09 only.

WhCosts

Column	Description
Plant_Code	Warehouse / plant ID (PLANT01–PLANT19)
Cost_Per_Unit	Storage cost in \$/unit

WhCapacities

Column	Description
Plant_Code	Warehouse / plant ID
Daily_Capacity	Max orders processable per day

ProductsPerPlant

Column	Description
Plant_Code	Warehouse / plant ID (includes outlier code CND9 in one row)
Product_ID	Product that plant can supply

VmiCustomers

Column	Description
Plant_Code	VMI-restricted plant (PLANT02, 06, 10, 11)
Customer	Customer that plant is allowed to serve

PlantPorts

Column	Description
Plant_Code	Warehouse / plant ID
Ports	Port the plant may ship through

Modeling tips (from dataset authors)

1. Demand side: Product_ID, Dest_Port, Unit_Quant, Weight from OrderList.
2. Cost side: WhCosts (storage) + FreightRates (shipping bands / min cost / rate).
3. Feasibility: PlantPorts, ProductsPerPlant, VmiCustomers, WhCapacities.
4. Do not expect FreightRates coverage for discontinued CRF carrier V44_3.

Files produced from this conversion

File	Format
wiki/sources/brunel-supply-chain-logistics-dataset/SupplyChainLogisticsProblems	Handwritten (this file)
wiki/sources/brunel-supply-chain-logistics-dataset/SupplyChainLogisticsProblems	Plain text
wiki/sources/brunel-supply-chain-logistics-dataset/SupplyChainLogisticsProblems	PDF
raw/sources/brunel-logistics-dataset/SupplyChainLogisticsProblems	Original Excel (this file)